CLINICAL TRIAL FAILURES · MISSED ENDPOINTS · ONCOLOGY PHASE 2,3 · STARTS 2015–2024

# TIGIT + PD-(L)1: 11 of 12 completed trials missed their primary endpoint

**11 completed trials across 3 sponsors and 11 molecules ran to the end and missed their primary endpoint on the sponsor's own posted result; 1 met it.** Only 4 of 59 closed trials were stopped early for efficacy or safety, which is why this class is read through its completed trials rather than a stop rate. Most completed trials post nothing readable, so the count is a floor.

**11 of 12**
completed trials with a readable primary result missed it · 1 met it

**3 / 11**
sponsors / molecules behind the misses

**6.8%**
stopped early: 4 of 59 closed trials (95% CI 2.7%–16.2%)

**46.1%**
of 128 trials have closed · 69 still open or unresolved

**11 of 12** **Completed and missed the primary endpoint.** Of 40 completed trials in this segment, 12 posted a primary result on ClinicalTrials.gov that can be read — the sponsor's own comparison held to the sponsor's own threshold, or its statement that the endpoint was missed: 11 missed and 1 met. Counted separately and never in the rate above; a floor, not a rate. 2 of the 31 completed trials that finished more than a year ago have posted no results at all — a gap in what can be known, not a sign of failure.

## The completed trials that missed

| Trial | Ph | Sponsor | Experimental drugs | Started | What the posted result says |
|---|---|---|---|---|---|
| NCT05116202 | 2 | Roche | TOBEMSTOMIG, ATEZOLIZUMAB, TIRAGOLUMAB | 2022-02 | Pathologic Response Rate (pRR) for Cohort 1 as Determined by Independent Pathologic Review — Difference in pRR 2.73 (95% CI -22.25 to 27.7), no p-value posted and no threshold stated: the interval is consistent with no effect under any conventional threshold, so not significant. |
| NCT05014815 | 2 | BeOne Medicines | OCIPERLIMAB | 2021-11 | Progression-free Survival (PFS) — one-sided p 0.4698, no threshold stated; 0.2 or above, so not significant at any one-sided threshold in use. Hazard Ratio (HR) 0.99 (95% CI 0.74 to 1.33). Method: Log Rank. |
| NCT05007106 | 2 | Merck & Co. | PEMBROLIZUMAB, VIBOSTOLIMAB, LENVATINIB, FLUOROURACIL | 2021-09 | Objective Response Rate (ORR) Per Response Evaluation Criteria in Solid Tumors (RECIST) 1.1 as Assessed by Blinded Independent Central Review (BICR) in Cohort A1 — one-sided p 0.2215, no threshold stated; 0.2 or above, so not significant at any one-sided threshold in use. Difference in Percentage 4.5 (95% CI -7.2 to 16.3). Method: Miettinen & Nurminen method. |
| NCT04725188 | 2 | Merck & Co. | PEMBROLIZUMAB, VIBOSTOLIMAB | 2021-04 | Progression Free Survival (PFS) Per Response Evaluation Criteria in Solid Tumors Version 1.1 (RECIST 1.1) by Blinded Independent Central Review (BICR) Assessment — p 0.9622, no threshold stated; 0.2 or above, so not significant under any conventional threshold. Hazard Ratio (HR) 1.4 (95% CI 0.96 to 2.02). Method: Regression, Cox. |
| NCT04732494 | 2 | BeOne Medicines | OCIPERLIMAB | 2021-03 | Objective Response Rate (ORR) Assessed by the Investigator — p 0.2114 against the sponsor's stated threshold of 0.05: not significant. Risk Difference (RD) 9.9 (95% CI -5.4 to 25.3). Method: Cochran-Mantel-Haenszel. |
| NCT04665843 | 2 | Roche | TIRAGOLUMAB | 2021-03 | Percentage of Participants With Confirmed Objective Response, as Determined by the Investigator Using Response Evaluation Criteria in Solid Tumors Version 1.1 (RECIST v1.1) — p 0.3886, no threshold stated; 0.2 or above, so not significant under any conventional threshold. Odds Ratio (OR) 1.57 (95% CI 0.56 to 4.41). Method: Cochran-Mantel-Haenszel. |

The 6 most recently started of 11 completed trials that missed. The diligence report lists every one, with the numbers or the sentence each verdict was read from.

## How this compares

TIGIT + PD-(L)1



6.8%

4/59 closed

PD-(L)1 combinations without TIGIT



5.2%

112/2148 closed

All oncology Phase 2,3



4.5%

355/7935 closed

**What is in this class:** 127 distinct experimental drugs across 128 trials (antibody 122, small molecule 37, adc 14, bispecific antibody 9). Most tested: TIRAGOLUMAB (47), ATEZOLIZUMAB (32), PEMBROLIZUMAB (19), RILVEGOSTOMIG (17), VIBOSTOLIMAB (16). A class groups drugs by what they act on; check that the grouping is one you would make before reading the rate as a property of the mechanism.

**How much does this rest on one decision?** The 4 stops came from 3 sponsor-asset programmes across 3 sponsors; the largest (Roche) contributed 2; removing that programme's trials from both sides leaves 4.8%. Screened alongside 85 other segments, the rate is within what screening that many classes could produce (q=1); the molecules and trials that failed here are the record either way. A further 9 closed trials here were terminated with no cause recorded in the registry; if every one of them were biological the rate would be 22.0%. Read it as a screen worth checking against the underlying trials.

Bars show the share of closed trials stopped for biological reasons. Confidence intervals overlap where sample sizes are small — read the counts, not just the bars.

## By start cohort

| Trials started | Stops | Closed | Rate | 95% CI |
|---|---|---|---|---|
| 2018–2020 | 0 | 21 | 0.0% | 0.0%–15.5% |
| 2021–2024 | 4 | 34 | 11.8% | 4.7%–26.6% |

## Where the stops sit

Efficacy / futility: 4.
Sponsors with most stops: Roche (2), BeOne Medicines (1), Merck & Co. (1).

## The stopped trials

| Trial | Ph | Sponsor | Experimental drugs | Stopped | Type | Registry stop reason |
|---|---|---|---|---|---|---|
| NCT06267001 | 3 | Roche | TIRAGOLUMAB | 2025-12 | Efficacy | Decision to terminate study was driven by results from preceding tiragolumab study (SKYSCRAPER-06;NCT04619797) \& sponsor's strategic priorities, not due to safety risks or quality reasons for any of molecules/combinations investigated in this study. |
| NCT04746924 | 3 | BeOne Medicines ONC | OCIPERLIMAB | 2025-05 | Efficacy | This decision was conducted by the sponsor based on lack of efficacy and not driven by safety concerns. |
| NCT05226598 | 3 | Merck & Co. MRK | PEMBROLIZUMAB, VIBOSTOLIMAB | 2024-09 | Efficacy | MK-7684A-007 was terminated early due to futility. A non-submission CSR, not intended to support any registration, was issued; therefore, an abbreviated CSR was generated. |
| NCT04308785 | 2 | Roche | TIRAGOLUMAB | 2023-07 | Efficacy | The sponsor's decision was based on the negative results of SKYSCRAPER-02. |

**Method**
Denominator: ClinicalTrials.gov interventional Phase 2,3 oncology trials started 2015–2024 that have closed (completed or terminated). Numerator: terminated trials whose registry stop reason is classified as biological (efficacy, safety or benefit–risk) by Classification V2 — held-out precision 95.5%, recall 95.3% (n=600). Drugs are linked to ChEMBL and the NCI Thesaurus; 76% of industry oncology trials in this window carry a resolved drug target, and a trial without one cannot enter a mechanism class. Intervals are Wilson 95%.

**Limits**
Not a failure rate: completed trials that missed their endpoints are not in the rate. They are counted separately where the sponsor posted a primary result that can be read — most did not, so that count is a floor — and programmes discontinued after a completed trial do not appear. Stop reasons are sponsor-reported. This is a closed-trial proportion, not a time-to-event analysis: only 46.1% of trials in this segment have closed, and a trial that stops early enters the denominator sooner than one that runs to completion, which can inflate the rate in immature segments. Recent cohorts have fewer closed trials, so their rates are less stable. Research signals, not clinical or investment advice.

**The whole cohort, trial by trial.** The diligence report names every trial behind these figures — the ones still running and the ones that completed and missed their primary endpoint included — and places your molecule against each one that failed. Free during the beta, in exchange for a work email, delivered immediately: **clinicaltrialfailures.com/asset-check**

Sources: ClinicalTrials.gov (NLM); ChEMBL (EMBL-EBI, CC BY-SA 3.0); NCI Thesaurus (NCI); RxNorm/RxClass (NLM); SEC EDGAR. Classification, linkage and rates by Clinical Trial Failures. Rebuilt weekly; this brief covers trials started 2015–2024.